

Data-Aware Cost Accounting and Hybrid Route Consolidation for Pharmaceutical Cold-Chain Logistics

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Abstract

Problem definition: Pharmaceutical cold-chain distribution couples routing decisions with refrigeration, vehicle utilization, service reliability, and data-quality constraints. We study anonymized dispatch and terminal records from a logistics case dataset. **Methodology/results:** A six-component accounting model is combined with daily route consolidation using Clarke–Wright initialization, genetic search, and 2-opt improvement. Across 735 dispatch records, modeled total cost is 5.4251 million CNY. Under the stated assumptions, consolidation changes modeled dispatches from 759 to 182 and modeled cost from 5.2088 to 1.1813 million CNY; the 77.3% difference is a model comparison, not an observed field saving. Twenty runs of the final search configuration have a reported coefficient of variation of 1.79%. A cross-file audit finds 92,009 common shipment identifiers and substantial address-quality limitations in SH9. **Managerial implications:** Optimization gains should be interpreted jointly with data quality. Incomplete addresses, approximate time-window constraints, and incompatible load definitions—including a recorded average load factor of 143.60%—must be resolved before capacity or savings claims are used operationally.

Keywords: cold-chain logistics; vehicle routing; route consolidation; cost accounting; data quality; pharmaceutical distribution

1 Introduction

Pharmaceutical distribution is an industrial-engineering problem in which transportation cost cannot be separated cleanly from product handling, temperature control, service reliability, and the quality of operational data. A route that appears inexpensive on distance alone may create refrigeration time, excessive detours, poor vehicle utilization, or service risk. Conversely, a sophisticated optimization algorithm cannot compensate for corrupted delivery addresses or incompatible capacity definitions.

Vehicle routing has developed from the classical truck-dispatching formulation of Dantzig and Ramser (Dantzig and Ramser, 1959) and the savings heuristic of Clarke and Wright (Clarke and Wright, 1964) into a large family of capacitated, time-window, heterogeneous-fleet, and environmentally aware models (Solomon, 1987; Braekers et al., 2016). Modern metaheuristics commonly combine population-based exploration with local improvement; examples include adaptive large-neighborhood search (Pisinger and Ropke, 2007) and hybrid genetic search with explicit diversity management (Vidal et al., 2013). Environmental extensions further show that distance, fuel use, vehicle load, and emissions need not move in exactly the same direction (Bektas and Laporte, 2011; Demir et al., 2014).

The present study is motivated by a practical case dataset rather than a benchmark instance. It contains trunk-dispatch records and a substantially larger terminal-delivery table, allowing three connected questions to be examined: (i) which cost components dominate the modeled baseline; (ii) how much route consolidation is available under explicit capacity and detour controls; and (iii) whether the terminal data are sufficiently reliable to support operational interpretation. The contribution is therefore a data-aware workflow rather than a claim of a new state-of-the-art vehicle-routing algorithm. The same records are used to connect cost accounting, route optimization, sensitivity analysis, repeated-run robustness, and cross-file data auditing.

Three distinctions are maintained throughout. First, all monetary savings are model-based comparisons rather than audited realized savings. Second, the terminal load factor above 100% is treated as a capacity/data-definition warning, not as an efficiency gain. Third, address reconstruction for the SH9 warehouse is partial, so the resulting fleet range is presented as an approximate planning estimate rather than a precise requirement.

2 Data and Problem Setting

2.1 Operational records

The analysis uses anonymized records supplied with a logistics modeling case. The trunk-dispatch table contains 735 dispatch records for the cost-accounting analysis. A terminal-delivery table contains 222,378 records for the operational indicators. The two operational tables can be partially linked through shipment identifiers; this linkage becomes important because some terminal addresses are collapsed to a common warehouse address.

The records do not provide all quantities required by a full operational deployment model. In particular, intra-day timestamps are insufficient for an exact vehicle-routing problem with time windows (VRPTW), several cost coefficients are engineering assumptions rather than audited company values, and nominal vehicle capacity is not always consistent with the recorded shipment quantity. The model therefore uses explicit approximations and reports their consequences rather than treating them as measured facts.

2.2 Three-stage workflow

The workflow has three stages. Stage 1 calculates a per-dispatch cost using fixed vehicle use, distance-dependent operation, refrigeration, utilization, waiting, and loading/unloading labor. Stage 2 consolidates daily customer visits subject to vehicle capacity, route structure, a maximum of four customer stops, and a route-elongation control. Stage 3 examines terminal load, punctuality, carbon accounting, and cross-file address recovery. This sequence links optimization to the data conditions under which the resulting recommendations would have to be implemented.

3 Cost Accounting Model

For vehicle type v , let P_v be purchase price, r the residual-value rate, N_{dep} depreciation years, N_{work} annual working days, I_v insurance, A_v inspection cost, R_v annual road-related fee, W_v daily labor, and M_v, F_v

monthly maintenance and violation costs. The daily fixed vehicle cost is modeled as

$$C_{fix}^v = \frac{P_v(1-r)}{N_{dep}N_{work}} + \frac{I_v + A_v + R_v}{N_{work}} + W_v + \frac{M_v + F_v}{30}. \quad (1)$$

Mileage-dependent cost uses Haversine distance d_h , a road-circuitry factor $\lambda = 1.3$, fuel cost c_{fuel}^v , and toll cost c_{toll}^v ,

$$C_{var}^v = (c_{fuel}^v + c_{toll}^v)\lambda d_h. \quad (2)$$

Four additional terms account for cold-chain and operating conditions. Refrigeration is approximated by a 5.0 kW unit at 1.5 CNY/kWh. Loading and unloading use two workers for two hours at 50 CNY per worker-hour, or 200 CNY per operation. For long trips above 1000 km, away-from-base waiting is valued at 30% of daily fixed cost. Vehicle utilization $\eta = q/Q_v$ is penalized by the smooth piecewise function

$$C_{pen}^v(\eta) = \begin{cases} 2C_{fix}^v(0.5 - \eta)^2, & \eta < 0.5, \\ 0, & 0.5 \leq \eta \leq 1, \\ 0.4C_{var}^v(\eta - 1)^2, & \eta > 1. \end{cases} \quad (3)$$

Both the function value and its first derivative are continuous at $\eta = 0.5$ and $\eta = 1$. For dispatch j , total modeled cost is the sum of fixed, distance-dependent, refrigeration, utilization, waiting, and loading/unloading terms.

Across the 735 records, the modeled total is 5.4251 million CNY. Table 1 gives the decomposition. Fixed and mileage-dependent terms account for 73.6% of the total, motivating the route-consolidation stage.

Table 1: Modeled cost decomposition for 735 dispatch records.

Component	Cost (10,000 CNY)	Share
Fixed vehicle cost	153.39	28.3%
Mileage-dependent cost	245.85	45.3%
Away-from-base waiting	31.18	5.7%
Refrigeration	21.56	4.0%
Loading/unloading	28.28	5.2%
Utilization penalty	62.25	11.5%
Total	542.51	100%

A one-factor-at-a-time sensitivity analysis varies fuel price, electricity price, labor rate, depreciation period, and annual working days. Raising fuel price from 6 to 10 CNY/L increases modeled total cost by 15.2%, corresponding to the largest reported normalized sensitivity ($S_{oil} = 0.76$). Electricity and labor have moderate reported sensitivities near 0.15, while depreciation period and working days have absolute sensitivities below 0.05. The exercise identifies influential assumptions but is not a joint uncertainty analysis.

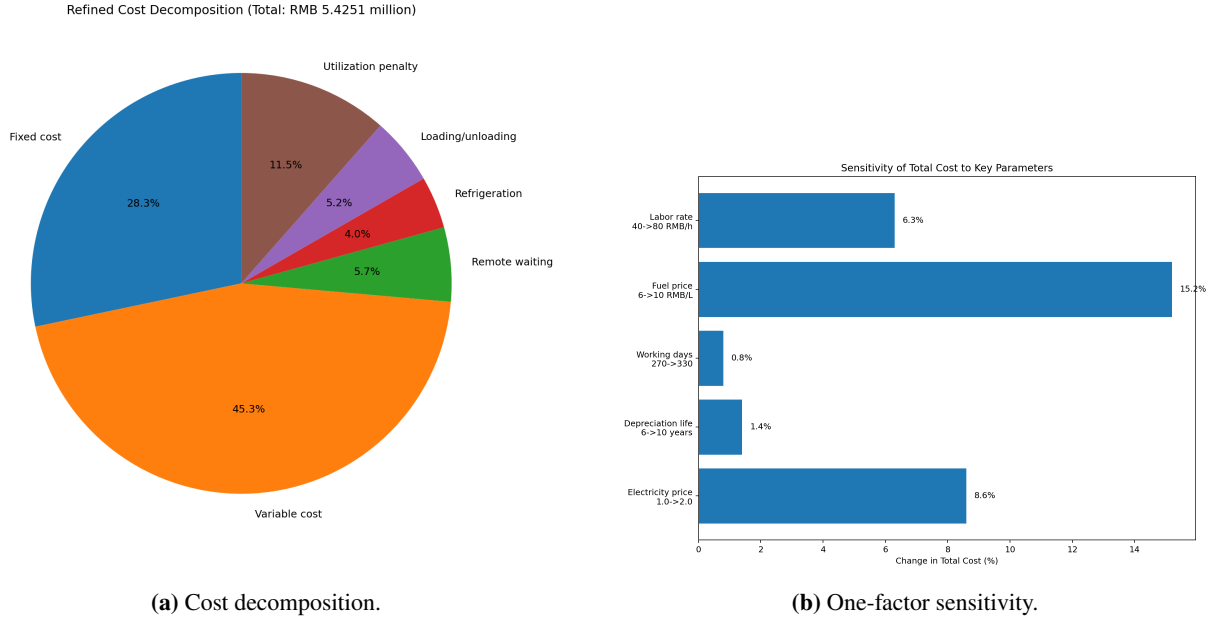


Figure 1: Cost-accounting diagnostics.

4 Route-Consolidation Model and Search Procedure

4.1 Routing formulation

For each day d , let N_d be the customer set and node 0 the depot. Binary x_{ijk} indicates whether vehicle k travels from customer i to j , y_{ik} assigns customer i to vehicle k , and u_{ik} is an MTZ auxiliary variable. The objective combines travel cost, fixed vehicle use, and a detour-control term,

$$\min Z = \sum_{k,i,j} c_{ij}^k x_{ijk} + \sum_k C_{fix}^k t_k + \sum_k \alpha s_k. \quad (4)$$

The constraint set requires each customer to be served once, load not to exceed nominal capacity, flow conservation, subtour elimination, and at most four customer stops per route. The four-stop limit is only a proxy for the stated 09:00–17:00 service window because the records do not contain sufficiently detailed intra-day timing for an exact VRPTW formulation. A route-distance condition also limits consolidated route length relative to the sum of independent depot round trips.

The route-elongation coefficient is evaluated over $\alpha \in \{1.2, 1.3, 1.4, 1.5, 1.6, 1.8, 2.0\}$. The recorded experiment selects $\alpha = 1.4$: it has the lowest modeled cost in the sweep (1.1813 million CNY) while the number of manually flagged implausible merges remains low. Looser settings reduce vehicle count slightly but introduce more long-detour combinations.

Table 2: Recorded route-elongation parameter sweep.

α	Cost (10,000 CNY)	Dispatches	Saving	Flagged merges
1.2	128.5	210	75.3%	0
1.3	122.3	195	76.5%	1
1.4	118.13	182	77.3%	3
1.5	118.5	180	77.2%	8
1.6	120.2	178	76.9%	15
1.8	124.8	175	76.0%	35
2.0	130.1	172	75.0%	55

4.2 Hybrid search

The constructive stage uses Clarke–Wright savings

$$S_{ij} = d_{0i} + d_{0j} - d_{ij} \quad (5)$$

and greedily merges positive-saving pairs subject to capacity, the four-stop limit, a 500 km pair-distance screen, and the route-elongation rule. This structured solution seeds a genetic search rather than using a wholly random initial population.

The final recorded configuration uses a population of 120 over 250 generations. Eighty percent of the initial population is formed by perturbing savings-based solutions and 20% is random. Tournament selection has size 3, order crossover probability is 0.85, total mutation probability is 0.15 across swap/insertion/inversion operators, decoded routes receive 2-opt improvement, and the best 10% are retained. This type of hybridization is consistent with the broader vehicle-routing literature, where population exploration is commonly combined with local improvement (Vidal et al., 2013).

The parameter setting was selected empirically. An earlier configuration with population 50 and 100 generations had a reported 20-run coefficient of variation of 8.3%; increasing search effort reduced the coefficient of variation to 1.79%. The recorded implementation reports approximately 5–8 minutes for a typical day with about 100 customer nodes on an Intel i7-10700, while a later tuning note reports runs up to roughly 15 minutes. We retain this as an implementation-dependent range rather than a hardware-independent performance claim.

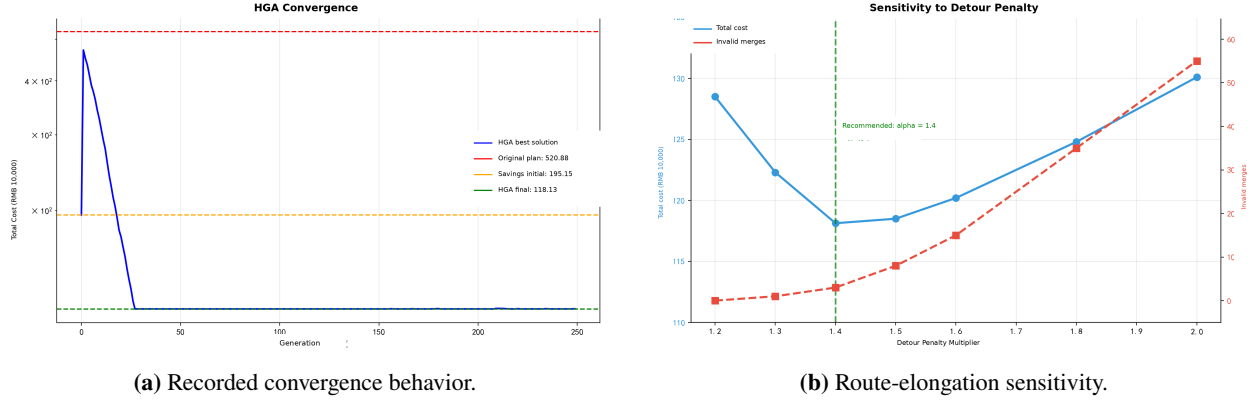
5 Computational Results

Under the stated model, the single-destination baseline has modeled cost 5.2088 million CNY and 759 dispatches, whereas the selected consolidated plan has modeled cost 1.1813 million CNY and 182 dispatches (Table 3). The difference is 4.0275 million CNY, or 77.3% relative to the modeled baseline. It should not be read as an observed post-deployment saving because road-network travel times, exact time windows, vehicle heterogeneity, and audited cost coefficients are not fully represented.

Twenty independent runs of the final configuration have a reported coefficient of variation of 1.79%,

Table 3: Modeled baseline and selected consolidated solution.

Plan	Cost (10,000 CNY)	Dispatches	Saving
Single-destination baseline	520.88	759	–
Consolidated plan	118.13	182	77.3%

**Figure 2:** Optimization diagnostics for the selected search design.

which indicates substantially more stable search behavior than the earlier 8.3% configuration. This statistic concerns algorithmic repeatability, not optimality: no exact lower bound or optimality certificate is available for the full case instance.

6 Terminal Operations and Data-Quality Audit

6.1 Operational indicators

The terminal table contains 222,378 records. The recorded indicators are an average load factor of 143.60%, a 17.5% share below 50% load, a 99.16% on-time rate under a 96-hour dispatch-to-receipt threshold, an average transport duration of 7 hours, and 157.73 t of modeled CO₂ emissions. These quantities should be interpreted jointly. In particular, an average load factor above 100% is either evidence of capacity violations or evidence that recorded shipment units and nominal vehicle capacity are not directly comparable. It is therefore a data/model diagnostic rather than a favorable utilization result.

Carbon is calculated from modeled travel distance, vehicle fuel consumption, and an emission factor of 2.68 kg CO₂/L. The analysis estimates that consolidating low-load movements could reduce 15–20% of ineffective mileage, corresponding to 23.7–31.5 t CO₂ under the stated assumptions. This is a scenario estimate, not a measured emissions reduction.

6.2 Cross-file linkage and SH9 address recovery

The strongest data-quality problem appears in warehouse SH9. In the terminal file, 33,302 SH9 records share a common receiving address, which collapses a naive geocoding pipeline to one point. A cross-file identifier

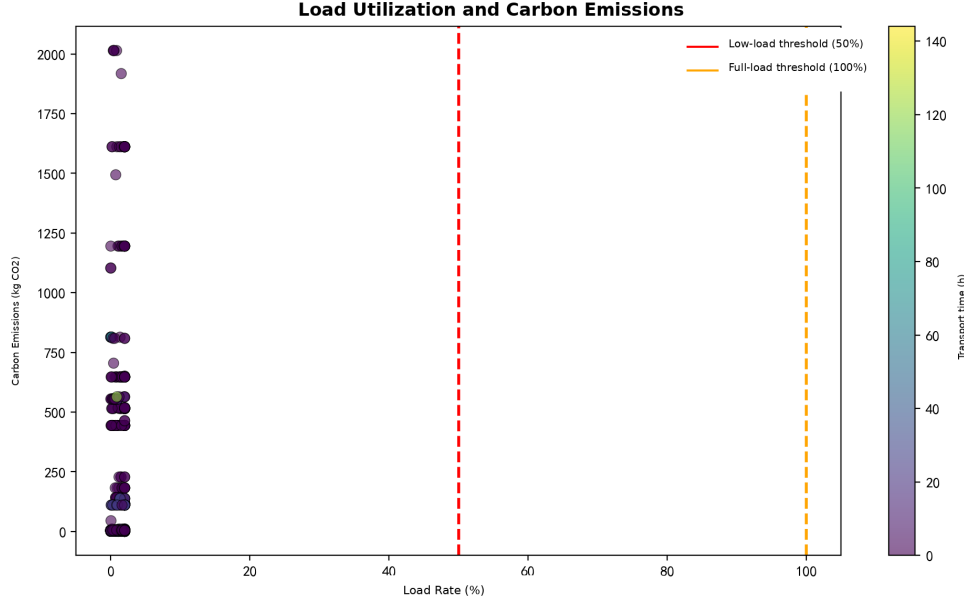


Figure 3: Recorded terminal-distribution diagnostics for load utilization and carbon-related performance.

audit finds 92,009 common shipment IDs overall and 13,580 common IDs for SH9, corresponding to 40.8% of SH9 records. These matched SH9 records yield 1,823 unique recovered addresses, of which 977 are in Shanghai and 846 are outside Shanghai.

The available analysis verifies address consistency on a sample of 2,622 common SH9 IDs from two daily files; all addresses in that validation sample agree across the two operational tables. We do *not* generalize this sample check into a claim of 100% agreement for every one of the 13,580 matched SH9 records.

For the matched subset, a transparent greedy nearest-neighbor routing procedure uses nominal capacity 100 items and a maximum service radius of 80 km, giving an approximate requirement of 400 vehicles. The unmatched majority is estimated from the spatial density observed in the matched records and then adjusted by same-area consolidation, dynamic vehicle sizing, and delivery-period clustering. Because the source analysis contains slightly different intermediate ranges, we retain only its conservative summary estimate of approximately 650–700 vehicles per week. The figure is a planning range, not a validated fleet requirement.

7 Discussion

7.1 What the optimization result establishes

The route-consolidation result establishes that, within the stated accounting and routing assumptions, the case data contain substantial redundancy relative to a single-destination dispatch baseline. The parameter sweep and repeated-run statistic provide two useful robustness checks: one tests sensitivity to the route-elongation rule, while the other tests stochastic search stability. Neither establishes global optimality or realized financial savings.

The cost decomposition explains why consolidation has such leverage. Fixed vehicle use and mileage-

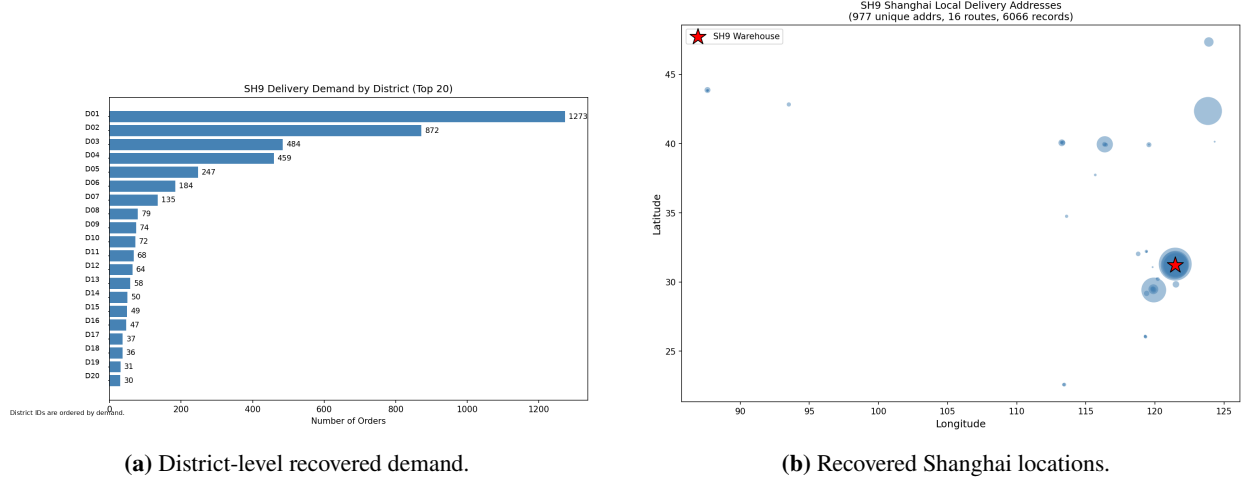


Figure 4: Spatial diagnostics for the SH9 address-recovery stage.

dependent operation account for nearly three quarters of the modeled total, so reducing unnecessary dispatches and mileage can dominate fine adjustments to refrigeration or labor coefficients. This mechanism is consistent with classical routing objectives, but the case adds cold-chain operating terms and explicit checks on data quality.

7.2 Relation to vehicle-routing literature

The method is intentionally simpler than state-of-the-art general-purpose VRP solvers. Adaptive large-neighborhood search and modern hybrid genetic search use richer neighborhood management, diversity control, and benchmark validation (Pisinger and Ropke, 2007; Vidal et al., 2013). The present contribution is not an algorithmic dominance claim. Instead, it demonstrates how a constructive heuristic, genetic exploration, local search, cost accounting, and data validation can be integrated in one applied decision-support pipeline.

Environmental interpretation should likewise remain modest. Pollution-routing research models fuel use and emissions as functions of load, speed, distance, and other operational factors (Bektas and Laporte, 2011; Demir et al., 2014). Here, the carbon metric is a simplified distance/fuel accounting calculation. It is useful for direction-of-change analysis but should not be treated as a high-fidelity emissions model.

7.3 Limitations and deployment requirements

Several limitations are material. First, Haversine distance multiplied by a fixed circuitry factor is not a road-network travel-time model. Second, the four-stop rule only approximates service windows; complete timestamps would be needed for a true VRPTW. Third, refrigeration power, waiting cost, and some operating thresholds are engineering assumptions rather than measurements from the focal company. Fourth, the SH9 address reconstruction observes only 40.8% of records directly and extrapolates the remainder. Fifth, the load-factor scale must be reconciled before capacity-based recommendations can be implemented.

A deployment study should therefore use complete geocoded destinations, road-network travel times, actual vehicle capacities in compatible units, explicit service windows and driver hours, audited cost coefficients, and

field validation of the proposed consolidated routes. Benchmarking against a strong ALNS or contemporary hybrid genetic search would also clarify how much of the reported saving comes from consolidation itself versus the particular heuristic implementation.

8 Conclusion

A data-aware pharmaceutical cold-chain workflow was developed around cost accounting, route consolidation, and terminal-data validation. The accounting model identifies mileage-dependent operation and fixed vehicle use as the dominant modeled cost components. Under the case assumptions, a Clarke–Wright-initialized hybrid genetic search reduces modeled dispatches from 759 to 182 and modeled cost from 5.2088 to 1.1813 million CNY, with a 20-run coefficient of variation of 1.79%. The data audit changes how those optimization results should be interpreted: an average load factor above 100%, collapsed terminal addresses, and incomplete SH9 linkage show that operational data quality is a first-order part of the optimization problem. The reported savings are therefore evidence of modeled consolidation potential, not a claim of realized company performance.

Data availability

The analysis is based on anonymized operational case records supplied with Problem B of the 2026 Huadong Cup University Mathematical Modeling Invitational. The manuscript reports aggregated values and author-generated diagnostic figures derived from those records. The public competition rules reviewed by the authors do not provide a blanket redistribution licence for the underlying case attachments; the raw files and organizer-supplied materials are therefore not redistributed with this submission. Any data sharing must comply with the organizer’s and data provider’s terms.

Acknowledgments and declaration of generative AI use

During preparation of this work, the authors used OpenAI ChatGPT (GPT-5.6 Sol) to assist with manuscript restructuring, English-language drafting and editing, and literature-search support in the Abstract and Sections 1–8, based on the authors’ research materials. The authors directed, reviewed, and edited the AI-assisted content, verified the scientific claims and references, and take full responsibility for the article. No generative-AI system was used to create or alter the underlying operational records or the reported optimization results.

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